Circulation: Cardiovascular Imaging

ADVANCES IN CARDIOVASCULAR IMAGING

Past, Present, and Future of Evaluating Diastolic Function

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ABSTRACT: The assessment of left ventricular diastolic function started with invasive pressure measurements and is currently primarily based on echocardiographic imaging. The current approach to the diagnosis of diastolic function relies on mitral inflow velocities, tissue Doppler early diastolic velocity of the mitral annulus, peak velocity of tricuspid regurgitation, pulmonary vein flow, left atrial size and function, and the noninvasive estimation of right atrial pressure. The recommended algorithm in the 2025 American Society of Echocardiography guidelines for estimation of mean left atrial pressure has been validated in a large multicenter study that included 951 patients. Further, that algorithm has shown incremental value to triaging clinical scores in achieving accurate heart failure with preserved ejection fraction diagnosis, against the invasive gold standard. Newer promising approaches have been developed over the past few years to study left ventricular diastolic function, including shear wave propagation velocity, left atrial conduit volume, and timing of mitral and tricuspid valve opening in the apical 4-chamber view. Artificial intelligence has been applied to this field as a rule-based decision tree algorithm, and as deep-learning neural networks to train models for diagnosing and grading diastolic dysfunction, and for diagnosing heart failure with preserved ejection fraction.

Key Words: artificial intelligence ■ atrial pressure ■ heart failure ■ pulmonary artery ■ pulmonary veins

The study of left ventricular (LV) diastolic function started with the measurement of right atrial (RA), right ventricular (RV), and pulmonary artery (PA) pressures. Patients with clinical evidence of left-sided heart failure had elevated RV and PA pressures in comparison with normal subjects. As to the first human catheterization, it was performed by the urologic surgeon, Dr Werner Forssmann, who introduced a ureteral catheter through the left antecubital vein into what he believed to be the RV.[1] Subsequently, Cournand and Richards reported on RA and RV pressure findings in patients with different pulmonary and cardiac pathology.[2] They noted elevated RV systolic pressure to 100 mm Hg in a patient with mitral stenosis and atrial fibrillation. With bed rest and the administration of cardiac glycosides, RV systolic pressure decreased to 40 to 60 mm Hg. In 1956, Dr Werner Forssmann received the Nobel Prize for Medicine or Physiology together with Cournand and Richards.[3] More direct evidence for elevated left atrial (LA) pressures can be obtained by using a PA or Swan-Ganz catheter which was reported on in 1970.[4] The catheter was a 5Fr double lumen catheter with a balloon proximal to its tip that is advanced into the pulmonary circulation under pressure monitoring and once in the distal branches of the PA, the balloon is inflated to record the PA occlusive pressure or the wedge pressure.

Later in 1976, Weiss et al[5] reported on the time course of the fall in isovolumic LV pressure in the isolated canine LV. This is the method most often used to derive the time constant of LV relaxation in invasive studies with high-fidelity pressure catheters. Another aspect of LV diastolic function is LV stiffness, which can be increased by an increase in LV mass as well as an increase in myocardial stiffness. To identify the underlying reasons behind increased stiffness, the modulus of chamber stiffness is derived and considered in the context of LV mass/volume ratio.[6] To obtain the modulus of chamber stiffness, LV pressures and volumes are measured using conductance catheters with varying preload as a balloon is inflated in the inferior vena cava. Thus, both the time

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Nonstandard Abbreviations and Acronyms

ASE	American Society of Echocardiography
DD	diastolic dysfunction
DT	deceleration time
E/A	ratio of mitral inflow peak early diastolic to peak atrial or late diastolic velocity
E/e'	ratio of mitral inflow peak early diastolic to mitral annulus early diastolic velocity by tissue Doppler
EDP	end diastolic pressure
HFpEF	heart failure with preserved ejection fraction
IVRT	isovolumic relaxation time
LA	left atrium
LAP	left atrial pressure
LARS	left atrial reservoir strain
LV	left ventricle
MR	mitral regurgitation
MVC	mitral valve closure
PA	pulmonary artery
PASP	pulmonary artery systolic pressure
PH	pulmonary hypertension
RA	right atrium
RAP	right atrial pressure
RV	right ventricle
TR	tricuspid regurgitation

constant of LV relaxation and the modulus of chamber stiffness are not accessible with routine left heart catheterization using fluid-filled transducers. Given these challenges, the search for noninvasive approaches to study LV diastolic function was the logical next step. The evolution of echocardiographic assessment will be discussed in this article, along with future developments in this field.

THE PAST

Early on, there was interest in the noninvasive assessment of LV diastolic function and LV filling pressures. Findings by M-mode, such as the B-bump in patients with elevated LV end diastolic pressure (EDP), were described.[7] Likewise, the time interval between the second heart sound by phonocardiography and opening of the mitral valve by echocardiography was used as an index of LA pressure (LAP) in patients with and without mitral stenosis.[8] The introduction of new technology fueled the ongoing development of novel indices of LV diastolic function. Using Pulsed wave Doppler echocardiography, mitral inflow and pulmonary venous flow into the left atrium are recorded. These signals are still relied on day to day for the assessment of LV diastolic function.

Mitral Inflow

Predominant early diastolic filling is seen in the presence of normal LV relaxation, whereas impaired relaxation leads to predominant late diastolic filling of the LV. Early diastolic filling could be assessed by mitral peak early diastolic velocity (E velocity), and late diastolic filling by mitral late diastolic velocity due to LA (hence atrial or peak A velocity) contraction. However, it was soon recognized that the inflow pattern is also affected by the LA-LV pressure gradient such that predominant early diastolic filling can occur in the presence of impaired LV relaxation when LAP is elevated. Conversely, predominant late diastolic filling can occur in the presence of normal LV relaxation when LAP is low. This was shown in animal models[9] and elegantly illustrated in humans by Appleton et al.[10]

There are other measurements that can be obtained from mitral inflow, including the acceleration time or the time between onset and peak E velocity (AT) and rate (mean acceleration rate given by peak E velocity/AT), as well as deceleration time (DT) and rate (deceleration rate given by peak E velocity/DT). The AT and rate are affected by LAP and LV relaxation, similar to how the mitral E velocity is affected. DT and rate are related to LV operating chamber stiffness,[11] such that in patients with increased chamber stiffness, DT is short and deceleration rate is increased. Given the dependence of mitral inflow pattern on LAP, the change in the inflow pattern from pseudonormal or restrictive in patients with heart failure after treatment to an impaired relaxation pattern is indicative of successful diuresis and lowering of LAP. Furthermore, a persistent restrictive filling pattern despite optimized guidelines directed medical therapy predicts worse outcomes and high mortality rates.[12]

Pulmonary Vein Flow

Initial reports related pulmonary venous flow systolic and diastolic velocities acquired by transesophageal echocardiography to mitral inflow velocities, mean wedge pressure and mean LAP.[13,14] Later, atrial velocity duration and the time difference between pulmonary vein atrial reversal wave duration and that of mitral A velocity duration were shown to be good indicators of LV EDP.[15] Pulmonary vein forward flow diastolic velocity into the LA has the same hemodynamic determinants as the mitral E velocity. Thus, the diastolic velocity is the highest velocity in young healthy subjects, and it decreases with age. The atrial velocity is not age dependent[12] and accordingly is a good surrogate of LV late diastolic pressures. In patients with elevated LAP, flow from the vein into the LA occurs more after LA emptying, which occurs during diastole after mitral valve opening. Therefore, the ratio of pulmonary vein flow (whether velocity or velocity time integral) during systole to flow during diastole is inversely related to mean LA pressure[13] and is an important finding to support the presence of elevated mean LAP. Further, the DT of the pulmonary vein diastolic velocity relates inversely with mean LAP and is a useful finding to draw inferences about LV filling pressure in patients with atrial fibrillation.[16]

Mitral Annular Velocities by Tissue Doppler

Tissue Doppler velocities are recorded by obtaining tissue signals with high amplitude and low frequency. Pulsed wave Doppler is used to obtain a spectral display of mitral annulus systolic, early (e'), and late (a') diastolic velocities. The e' velocity has received the most attention as an index of LV relaxation, with animal and human studies revealing its hemodynamic determinants.[17–19] Ratio of mitral inflow peak early diastolic to mitral annulus early diastolic velocity by tissue Doppler (E/e') ratio was introduced in 1997 as a dimensionless index of LV filling pressure.[19] Mitral E and annular e' velocities have a high feasibility of satisfactory acquisition. The E/e' ratio has acceptable accuracy for LV filling pressures estimation in several cardiovascular diseases. Both e' velocity and E/e' ratio are currently relied on for drawing conclusions about LV diastolic function. Notwithstanding, the E/e' ratio should not be relied on in patients with pericardial compression syndromes, mitral stenosis, patients with hemodynamically significant primary mitral regurgitation (MR) and normal LV ejection fraction, and moderate to severe mitral annular calcification.

Normally, LV long axis expands in early diastole before opening of the mitral valve, whereas in patients with impaired LV relaxation, mitral valve opening precedes LV long axis expansion.[20,21] The time delay between the onset of mitral valve opening and the subsequently occurring mitral annulus e' velocity has a direct relation with the time constant of LV relaxation[20,21] and is associated with worse outcomes in patients with primary MR and normal LV EF.[22]

Other Indices of LV Diastolic Function

Several indices have been introduced over the years as indices of LV relaxation, including color M-mode flow propagation velocity, diastolic strain rate during isovolumic relaxation, diastolic strain rate during early diastole, and untwisting rate.[23,24] Due to lower feasibility and higher measurement variability, these measurements are not recommended for the day-to-day assessment of LV diastolic function.

Previous Diastolic Function Assessment Guidelines

In 2009, the American Society of Echocardiography (ASE) and the European Association of Echocardiography developed the first guidelines document.[12] The recommendations included many signals with the expectation that these will be obtained and considered collectively in reaching conclusions about LV diastolic function and LAP. However, there was confusion and lower interobserver agreement in part related to the question of how to handle conflicting findings when several signals are considered together without a clear hierarchy.

In 2016, the second guidelines were published as a joint document by ASE and European Association of Cardiovascular Imaging. Two algorithms were recommended.[24] The first algorithm (algorithm A) to be applied to determine if LV diastolic function is normal or abnormal in patients with normal LV EF based on 4 variables: annular e' velocity, E/e' ratio, peak tricuspid regurgitation (TR) velocity, and LA maximum volume index. If the majority of the signals are indicative of diastolic dysfunction (DD) and also in patients with myocardial disease (defined by LV EF <50%, pathological LV hypertrophy, and abnormally reduced LV global longitudinal strain), the second algorithm (algorithm B) is applied to estimate LAP. There was good interobserver agreement for estimating LAP and grading LV diastolic function.[25] However, several cardiologists ended up calling indeterminate diastolic function based on relying only on the 4 signals above.[23] The challenges in clinical application, along with the emergence of data on LA strain, were the impetus behind developing ASE 2025 diastolic function guidelines recommendations.

THE PRESENT

The 2025 guidelines have important differences from the 2016 guidelines aimed at decreasing indeterminate calls and increasing the sensitivity of detecting patients with normal LVEF and elevated LAP. Table 1 shows these differences. ASE 2025 guidelines have specific recommendations for diagnosing LV DD, estimating LAP, and diagnosing heart failure with preserved ejection fraction (HFpEF).[26]

Diagnosis of LV DD

The diagnosis of DD relies on direct evidence of impaired LV relaxation and the consequences of increased myocardial stiffness, namely, elevated LAP, LA, and LV structural changes. Age-adjusted e' velocity is important for detecting impaired LV relaxation. However, particularly in normal hearts, e' velocity is affected not only by LV relaxation but also by suction and transmitral pressure gradient.[17,18] Accordingly, the correlation of e' velocity with the time constant of LV relaxation is not perfect, and hence the need for consideration of additional echocardiographic variables. Another indicator of DD is mitral inflow showing an impaired LV relaxation filling pattern with an ratio of mitral inflow peak early diastolic to peak atrial or late diastolic velocity (E/A) ratio that is lower than what is expected based on age, provided the patient

Table 1. Comparison Between 2016 and 2025 Diastolic Function Guidelines

ASE/EACVI 2016 guidelines	ASE 2025 guidelines
Guidelines depend on LV EF	One algorithm irrespective of LV EF
Four main variables in step I: mitral annulus e' velocity, E/e' ratio, LA maximum volume index, and peak TR velocity	In step I, 3 variables: average e' velocity, average E/e' ratio, and PASP or peak TR in the absence of pulmonary disease. In step II: LARS pulmonary vein S/d ratio, LA maximum volume index, and IVRT
LA maximum volume index included in the first step for evaluation of LV diastolic function	LA maximum volume index included in the second step, but not the first step, for LAP estimation
Higher rate of indeterminate LV diastolic function, and or indeterminate LAP	Very low rate of indeterminate LV diastolic function when all of the signals in point 2 are acquired
Lower accuracy overall, particularly lower sensitivity, negative predictive value, and accuracy in patients with normal LV EF	Higher accuracy overall, particularly higher sensitivity, negative predictive value, and accuracy in patients with normal LV EF
Diastolic stress echocardiography is recommended in symptomatic patients with grade I diastolic dysfunction	Diastolic stress echocardiography is recommended in symptomatic patients with grade I diastolic dysfunction

ASE indicates American Society of Echocardiography; D, diastolic; E, mitral inflow peak early diastolic velocity; e', mitral annulus early diastolic velocity; EF, ejection fraction; IVRT, isovolumic relaxation time; LA, left atrium; LAP, left atrial pressure; LARS, left atrial reservoir strain; LV, left ventricle; PASP, pulmonary artery systolic pressure; S, systolic; and TR, tricuspid regurgitation.

does not have dehydration with reduced LAP. Furthermore, in most patients with DD, elevation of mean LAP occurs, and hence the recommendation to consider indices of elevated LAP in the evaluation for the presence of DD. These include E/e' ratio, LA reservoir strain (LARS), and restrictive LV filling pattern. Over time, patients with chronic elevation in mean LAP, the LA enlarges, and the LA maximum volume index exceeds the upper limits of normal, and thus, LA size is an indicator of abnormally elevated mean LAP. In many patients with DD, LV hypertrophy (and at times concentric remodeling) is present, and an abnormally increased LV mass index is present. Importantly, LA enlargement and LV hypertrophy were relied on for the selection of patients with DD to include in the randomized controlled clinical trials studying the use of SGLT2 inhibitors in patients with HFpEF.[27] Thus, an abnormally reduced e' velocity in addition to any 1 measurement of the LA and LV structural and functional abnormalities described above is recommended for diagnosing DD. If e' velocity falls in the normal range, then a total of 2 LA or LV (2 LA, 2 LV, or 1 LA and 1 LV) measurements are recommended.

Estimation of Mean LAP

The 2025 ASE guidelines recommendation is based on a validated algorithm.[28] The assessment relies initially on echocardiographic signals with high feasibility, high interobserver agreement, and high specificity for elevated mean LAP. The 3 measurements are average age-adjusted e' velocity as an index of LV relaxation, average E/e' ratio, and PA systolic pressure (PASP) as indices of elevated mean LAP. If all 3 variables are in the abnormal range, then LV DD is present, and mean LAP is elevated. Mitral inflow and annular e' velocity have the highest feasibility of all echocardiographic measurements and, in many cases, can be adequately recorded even in patients with suboptimal 2-dimensional images.[28] Peak TR velocity and mean RA pressure (RAP) are needed for the estimation of PASP based on the modified Bernoulli expression. In labs that have a high rate of administration of agitated saline for the study of patients with pulmonary hypertension (PH), and ultrasound enhancing agents for endocardial border definition, recording a complete TR jet is possible with a feasibility that exceeds 80%.[28] PASP is a more reliable index of elevated LAP in comparison with peak TR velocity only. This can be seen in patients with RV systolic dysfunction, where peak TR velocity is <2.8 m/s but mean RAP is elevated in the 10-20 mm Hg range. The consideration of only peak TR velocity in these patients can lead to missing patients with elevated PASP and elevated mean LAP.

If e' velocity is reduced, whereas average E/e' ratio and PASP are in the normal range, then the next step is to examine the E/A ratio. An E/A ratio ≤0.8 indicates the presence of DD and normal LAP. If the combination of variables from step 1 is different from the 3 distinct possibilities of only e' velocity reduced, all 3 variables normal, all 3 variables abnormal, a second group of variables is examined. This includes pulmonary vein flow, LARS, LA maximum volume index, and isovolumic relaxation time (IVRT). The complete examination should include all 4 signals, as the presence of certain clinical or technical findings can lead to the exclusion of 1 or more signals from consideration (Table 2). It is important to emphasize that a complete examination for the assessment of LV diastolic function should contain all of the above signals to achieve low false-negative and low false-positive rates.

Of all 4 signals, the false positive rate is highest when only the increased LA volume maximum volume index is relied on for concluding the presence of elevated LAP. This is because the LA maximum volume index has the lowest correlation with LV filling pressures and does not detect acute changes in LAP.[29] Expectedly, confidence in reaching a reliable conclusion about LAP is highest when all the acquired signals are concordant in indicating the presence of normal or elevated LAP. There are also supplementary variables recommended

Table 2. Limitations for Variables Included in 2025 ASE Algorithm for LAP Estimation

E/e' ratio	Not reliable in pericardial compression syndromes, ≥moderate primary MR, moderate or severe MAC, any degree of mitral stenosis, surgical ring, M-TEER, prosthetic mitral valve, and normal subjects with acute changes in volume status
PASP and peak TR velocity	Not an index of LAP in the presence of noncardiac PH
LA maximum volume index	Anemia, heart transplant recipients with biatrial technique, hyperdynamic state, athletic status, ≥moderate primary MR, atrial flutter, and atrial fibrillation
LA Reservoir strain	≥Moderate primary MR, heart transplant recipients, atrial flutter, and atrial fibrillation
Pulmonary vein S/D velocity ratio	≥Moderate primary MR, heart transplant recipients, hypertrophic cardiomyopathy, atrial flutter, and atrial fibrillation
IVRT	In addition to LAP, IVRT is affected by heart rate and LV end systolic pressure.

ASE indicates American Society of Echocardiography; D, diastolic; E, mitral inflow peak early diastolic velocity; e', mitral annulus early diastolic velocity; IVRT, isovolumic relaxation time; LA, left atrium; LAP, left atrial pressure; LV, left ventricle; MAC, mitral annular calcification; MR, mitral regurgitation; M-TEER, mitral transcatheter edge to edge repair; PASP, pulmonary artery systolic pressure; PH, pulmonary hypertension; S, systolic; and TR, tricuspid regurgitation.

in the guidelines in case the available signals are not available or inconclusive (Table 3). Importantly, in a large multicenter study, the 2025 ASE guidelines resulted in only 2 of 951 cases being deemed indeterminate, along with good accuracy.[28] Figures 1 through 4 and Figure S1 show case examples of LAP estimation using the 2025 ASE guidelines.

Practical Aspects for the Successful Application of the 2025 Guidelines

The successful application of the 2025 guidelines starts with an imaging protocol that includes all the recommended measurements in the algorithm, which is needed to reach high sensitivity in identifying patients with elevated LAP. Most measurements in the 2025 guidelines have high feasibility, but LA strain and pulmonary vein acquisition can pose challenges for less experienced sonographers. The successful acquisition of pulmonary vein flow can be guided by color Doppler showing flow from the right paraseptal vein into the LA. A sample volume of 1 to 2 mm is then placed at 1 to 2 cm in the vein, and not at the junction of the vein with the LA. The feasibility of acquisition in outpatients is close to 90%, albeit it is more challenging to get satisfactory signals in ICU patients who are on mechanical ventilation and have limited access because of instrumentation. Feasibility can be further enhanced if ultrasound enhancing agent are administered, with careful adjustment of gain and compression settings to avoid overestimation of the true velocity.

For LA strain measurement, and also for correct LA volume measurement, it is important to obtain dedicated LA views, as the long axis of the LA is in a different plane from the long axis of the LV. This means one would acquire foreshortened apical views that are not suitable for LV volume measurements but are critical for LA volumes and LA strain measurements. This is usually obtained from an intercostal space that is higher than the space where the true apex is. It should be noted that foreshortened LA views decrease the sensitivity for detecting elevated LAP due to underestimation of LA maximum volume index and overestimation of LARS. In our laboratory, we routinely acquire the LA dedicated views and use the ultrasound platform to measure LARS. The sonographer checks whether there is good tracking, and if so, the file is stored for subsequent review by the interpreting physician. Our laboratory obtains LARS measurements in all patients with adequate 2-dimensional image quality and not on a selective basis, depending on whether LARS is needed for diastolic function assessment in a given case. This increases the experience and the confidence level of sonographers, without adding much time to the duration of image acquisition. It also

Table 3. Supplementary Variables for LAP Estimation

PR EDV*	PR EDV ≥2 m/s consistent with elevated LAP
PR EDP*	PR EDP ≥16 mm Hg consistent with elevated LAP
Pulmonary vein Ar	Ar >35 cm/s consistent with elevated LV EDP
Ar duration minus mitral A velocity duration (Ar–A)	Ar-A>30 ms is consistent with elevated LV EDP
Mitral inflow L velocity	L velocity ≥50 cm/s consistent with elevated LAP
Change in mitral inflow pattern with Valsalva	Change in E/A <50% consistent with elevated LAP
Premature termination of mitral inflow†	If present, LV EDP is elevated
Diastolic mitral regurgitation†	If present, LV EDP is elevated

Ar indicates atrial velocity; E/A, ratio of mitral inflow peak early diastolic to peak atrial or late diastolic velocity; EDP, end diastolic pressure; LAP, left atrial pressure; LV, left ventricle; and PR, pulmonary artery.

*In the absence of pulmonary parenchymal disease and the absence of pulmonary vascular disease.

†Can be considered after exclusion of PR interval >200 ms, second degree AV block, third degree AV block, atrial flutter, and atrial fibrillation with atrial contraction.

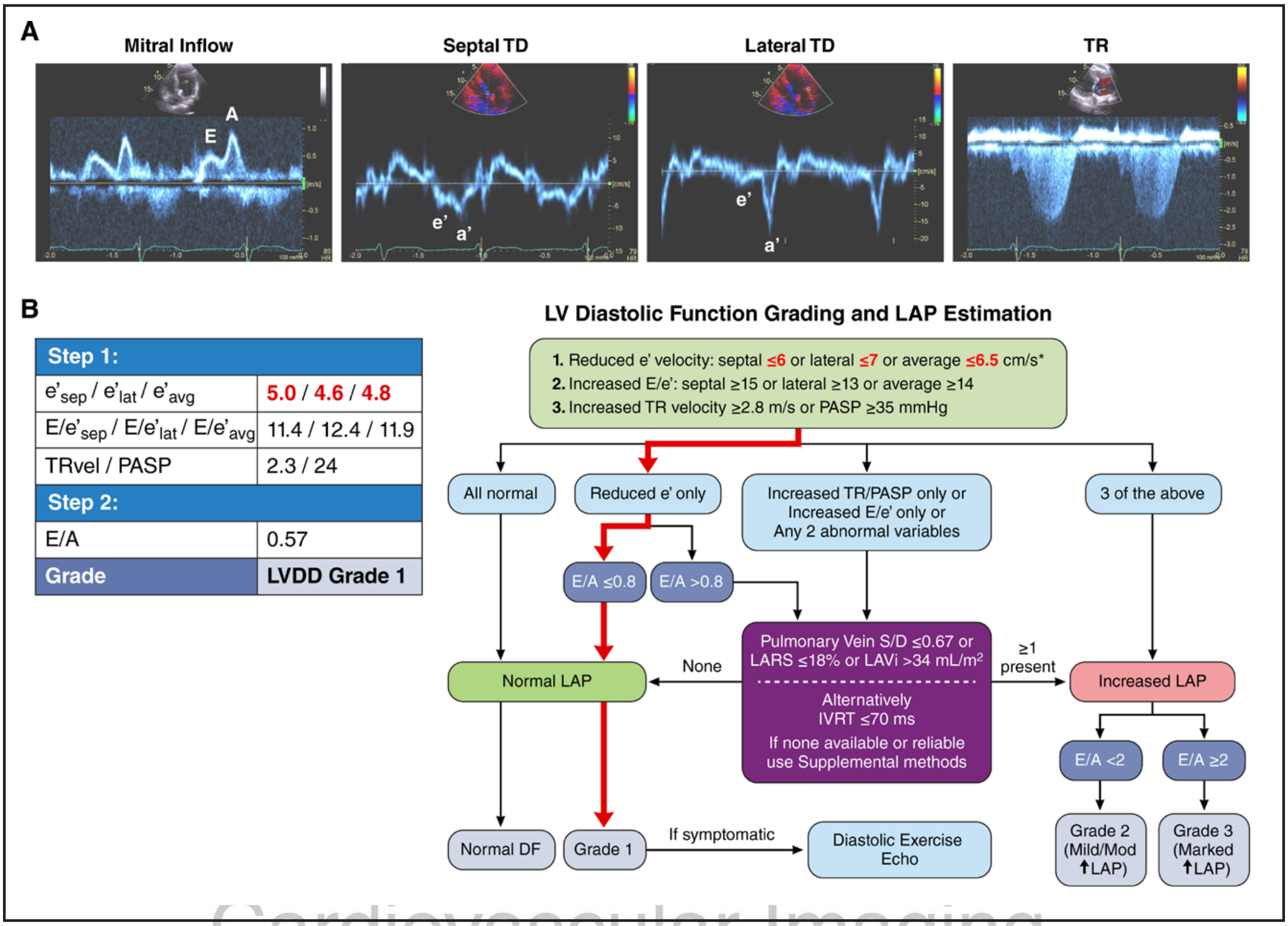

Figure 1. Left ventricular (LV) diastolic function assessment and left atrial pressure (LAP) estimation in an asymptomatic 86-year-old patient with hypertension.

A, Imaging findings from an 86-year-old male with hypertension who was asymptomatic at the time of the echocardiogram. LVEF was measured at 68%. LA maximum volume was 46 mL, and LA maximum volume index was normal at 24 mL/m². LV mass index was increased to 120 gm/m². Mitral inflow pattern (**left**) shows a ratio of mitral inflow peak early diastolic to peak atrial or late diastolic velocity (E/A) ratio of 0.57. Early diastolic velocity (e') at the septal (**center left**) and lateral (**center right**) mitral annulus is reduced with age, with an average ratio of mitral inflow peak early diastolic to mitral annulus early diastolic velocity by tissue Doppler (E/e') ratio of 11.9. Peak tricuspid regurgitation (TR) velocity is normal at 2.3 m/s (**right**), and the inferior vena cava (IVC) diameter was normal with spontaneous collapse with respiration (estimated right atrial pressure or RAP 3 mm Hg), yielding an estimated pulmonary artery systolic pressure (PASP) of 24 mm Hg. **B**, Step-by-step approach for applying 2025 American Society of Echocardiography guidelines to assess LV diastolic function and estimate mean LAP based on the findings shown in Figure 2A. Abnormal values are shown in red. IVRT indicates isovolumic relaxation time.

allows the interpreting physician to examine and consider a wide array of signals.

From an interpretation perspective, the interpreting team needs to consider the technical quality of each of the acquired signals and whether there are specific pathological or physiological conditions that invalidate the inclusion of that particular signal as the diagnosis of DD is formulated. For example, mitral inflow signals have to be at the level of tips using a 1 to 2 mm sample volume and with good alignment with mitral inflow, which can be guided by color Doppler. At times, off-axis views are obtained, resulting in suboptimal alignment with mitral inflow and underestimation of peak mitral E velocity. In these cases, mitral E velocity should not be measured. One of the most common mistakes is relying on an incomplete TR jet to draw inferences about LAP. If the jet is incomplete, it is reasonable to use intravenous saline or ultrasound enhancing agent in case they are used for endocardial border delineation (avoiding blooming signals, which can lead to overestimation of peak TR velocity). If the jet is incomplete and the signal is not enhanced, then the interpreting physician needs to go to the second step with its 4 variables.

In patients who have reasons for LA enlargement as athletes or end-stage liver disease, LA volume should not be relied on for drawing conclusions about LAP. Likewise, in normal healthy young adults, predominant diastolic flow in the pulmonary veins should not be used to conclude that LAP is elevated, as in these subjects, the predominant diastolic flow pattern in pulmonary veins is normal and occurs in the setting of normal LV diastolic function. For LARS measurements, one has to be careful

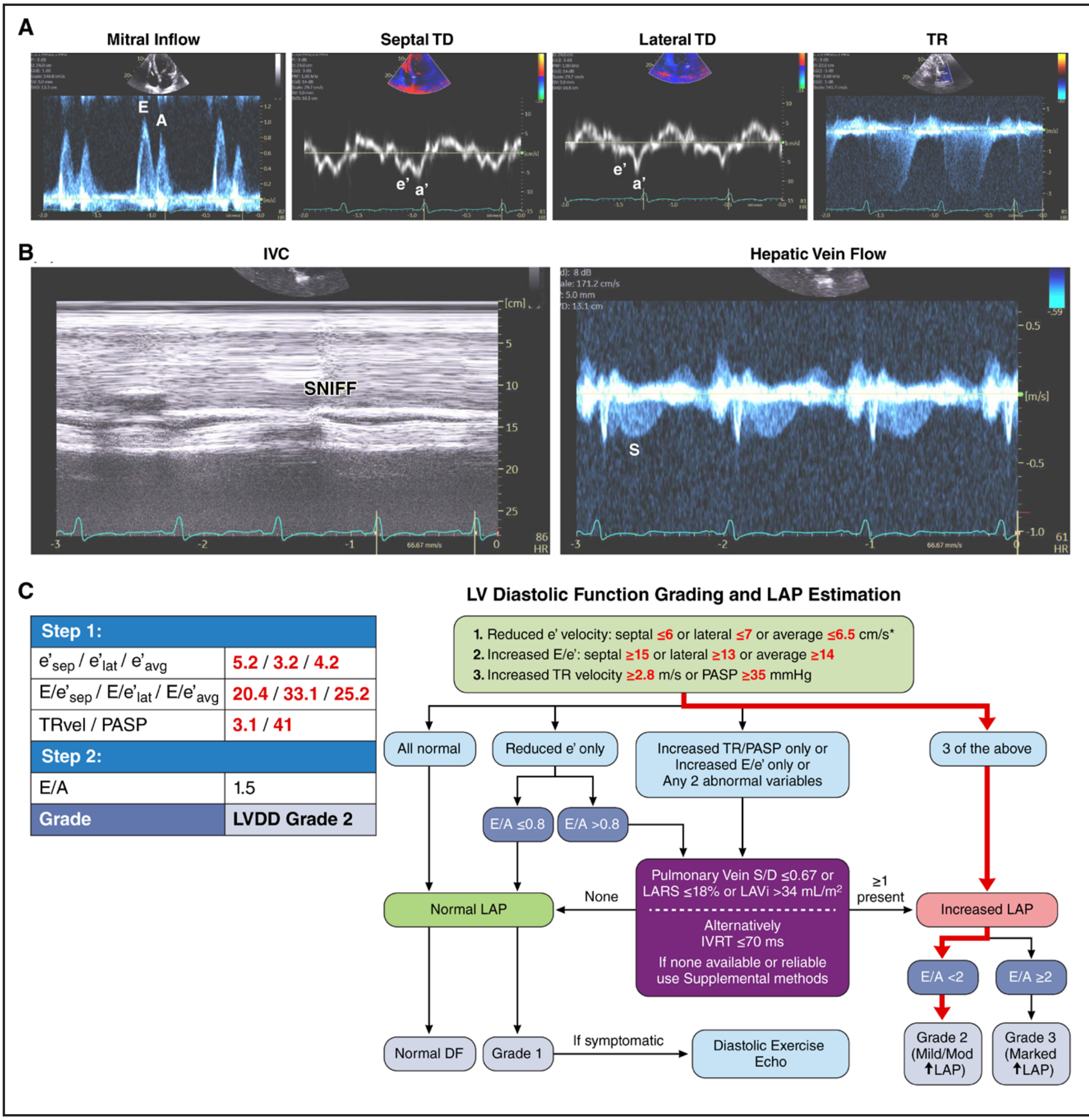

Figure 2. Left ventricular (LV) diastolic function assessment and left atrial (LA) pressure estimation in a 60-year-old patient with shortness of breath and New York Heart Association (NYHA) class III symptoms.

A, Imaging findings from a 60-year-old male with shortness of breath and NYHA class III symptoms at the time of the echocardiogram. LVEF was measured at 21%. LA maximum volume was 167 mL, and the LA maximum volume index was increased to 69 mL/m^2. LV mass index was increased at 135 gm/m^2. Mitral inflow (**left**) shows a ratio of mitral inflow peak early diastolic to peak atrial or late diastolic velocity (E/A) ratio of 1.5. Early diastolic velocities (e') at the septal (**center left**) and lateral (**center right**) mitral annulus are reduced beyond what is expected for age, with an average e' velocity of 4.2 cm/s and an average ratio of mitral inflow peak early diastolic to mitral annulus early diastolic velocity by tissue Doppler (E/e') ratio of 25.2. Peak tricuspid regurgitation (TR) velocity is increased at 3.1 m/s (**upper right**). **B**, Inferior vena cava (IVC) diameter is normal with collapse during sniffing (**left**). Hepatic vein flow (**right**) shows forward flow during systole (S). Both findings are consistent with normal mean right atrial pressure (RAP). The estimated pulmonary artery (PA) systolic pressure is, therefore, 41 mm Hg. **C**, Step-by-step approach for applying 2025 American Society of Echocardiography guidelines to assess LV diastolic function and estimate mean left atrial pressure (LAP) based on the findings shown in Figure 3A and 3B. Abnormal values are shown in red. LARS indicates left atrial reservoir strain.

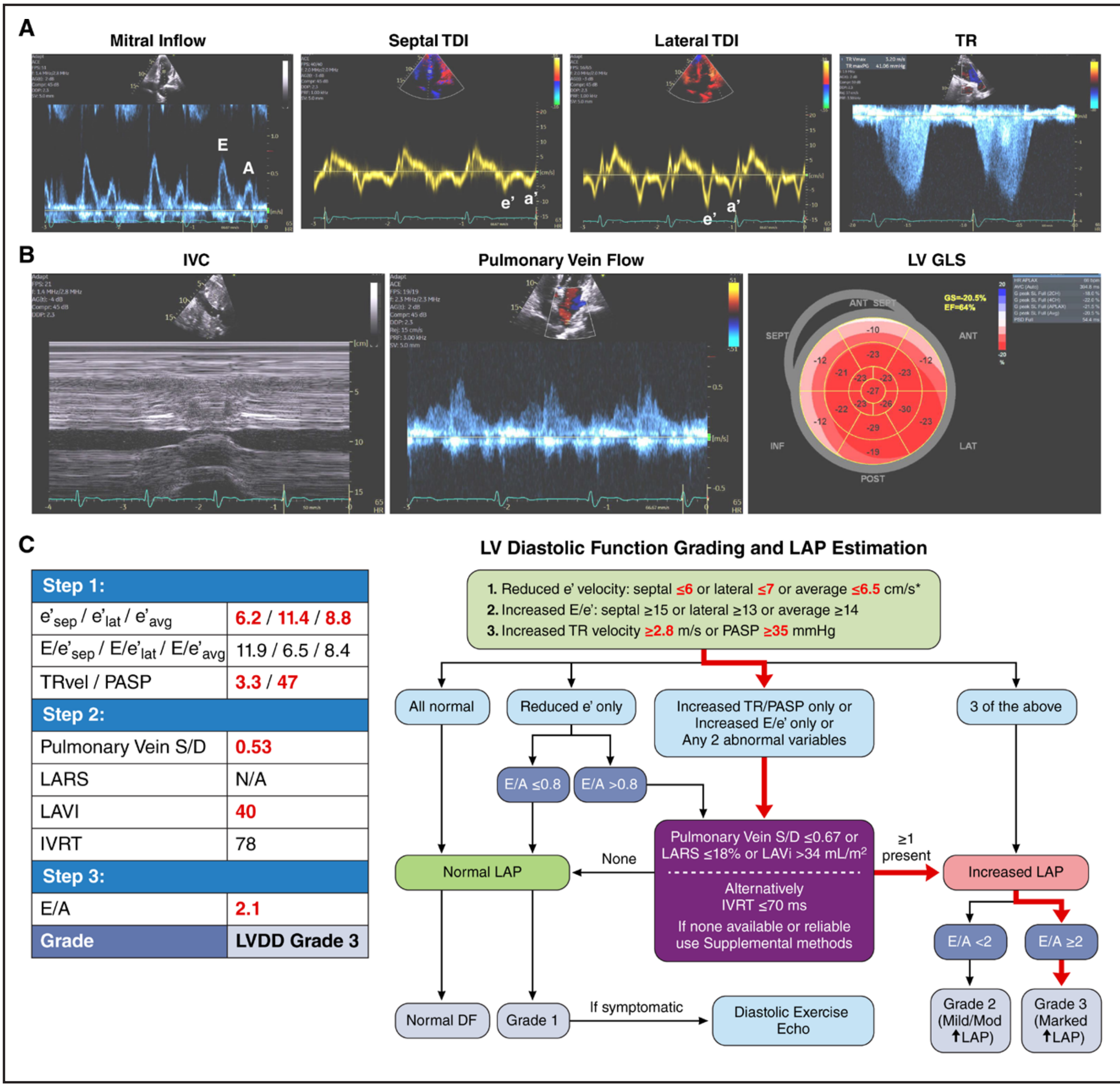

Figure 3. Left ventricular (LV) diastolic function assessment and left atrial (LA) pressure estimation in a 93-year-old patient with shortness of breath and chest pain.

A, Imaging findings from a 93-year-old female with shortness of breath and chest pain at the time of the echocardiogram. LVEF was measured at 64%. LA maximum volume was 71 mL, and the LA maximum volume index was increased to 40 mL/m². LV mass index was normal at 66 gm/m². Mitral inflow (**left**) shows an E/A ratio of 2.1. Early diastolic velocities (e') at the septal (**center left**) and lateral (**center right**) mitral annulus are normal for age, with an average e' velocity of 8.8 cm/s and an average E/e' ratio of 8.4. Peak tricuspid regurgitation (TR) velocity is increased at 3.3 m/s (**right**). **B**, Inferior vena cava (IVC) diameter is normal with collapse during sniffing (**left**), consistent with normal mean right atrial pressure (RAP). Pulmonary vein flow (**center**) shows reduced systolic (S) to diastolic (D) velocity ratio at 0.53, which is markedly abnormal for a 93-year-old patient and consistent with elevated mean left atrial pressure (LAP). The estimated pulmonary artery systolic pressure (PASP) is elevated to 47 mm Hg. LV global longitudinal strain (GLS; **right**) is normal at 20.5%. **C**, Step-by-step approach for applying 2025 American Society of Echocardiography guidelines to assess LV diastolic function and estimate mean LAP based on the findings shown in Figure 4A and 4B. Anormal values shown in red. LARS indicates left atrial reservoir strain.

in scrutinizing the presented value to ascertain adequate tracking in dedicated LA views and determine if the measurement makes sense. LARS reflects the change in LA volume due to LA filling during systole and corresponds to the difference between LA maximum and minimum volumes relative to the LA minimum volume at the end of diastole. When there is a large LA volumetric change in an average-sized atrium, the LARS value should be in the normal range. If a low value is obtained in this case, then one has to question the accuracy of the measurement and verify that indeed the correct value has been obtained.

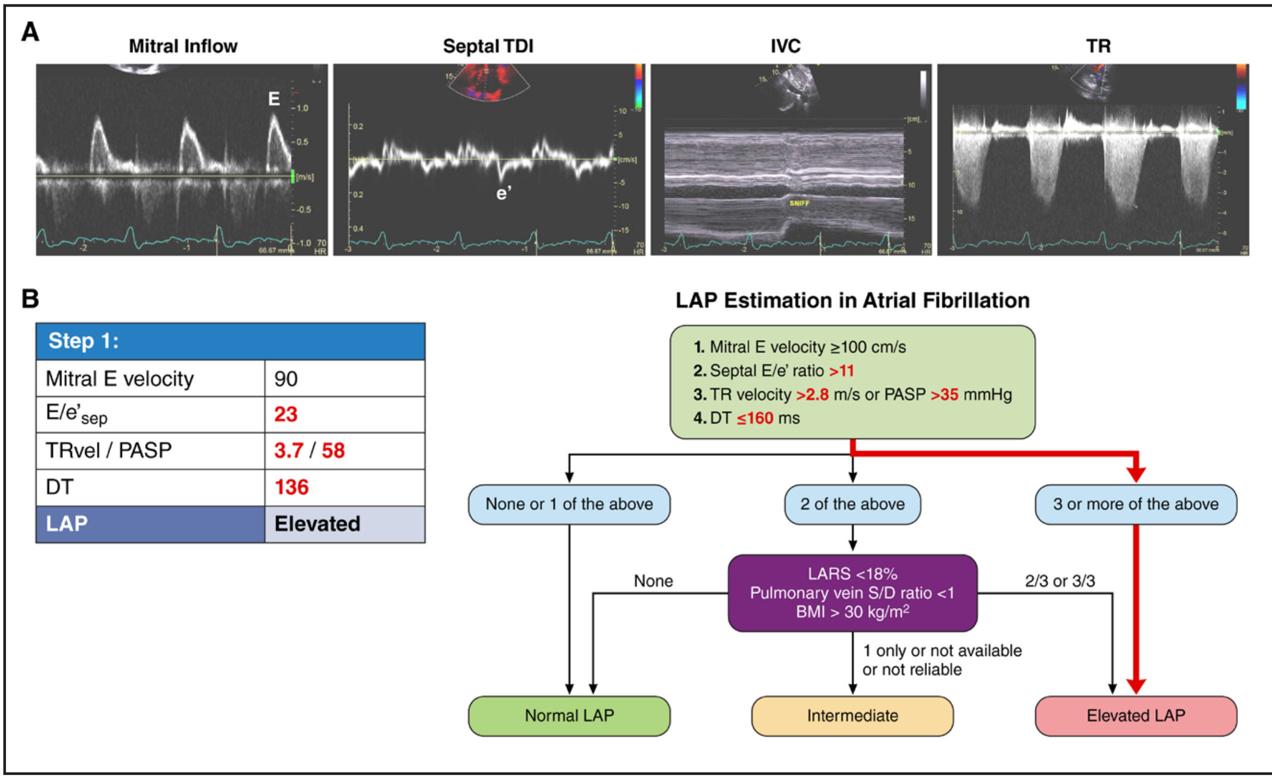

Figure 4. Left ventricular (LV) diastolic function assessment and left atrial (LA) pressure estimation in an 80-year old patient with atrial fibrillation.
A, Imaging findings from an 80-year-old female with heart failure with preserved ejection fraction (HFpEF) and atrial fibrillation, complaining of shortness of breath at the time of the echocardiogram. LVEF was measured at 66%. LA maximum volume was 107 mL, and the LA maximum volume index was increased to 64 mL/m^2. Mitral inflow (**left**) shows little beat-to-beat variation in peak E velocity with peak velocity at 90 cm/s and deceleration time (DT) of 136 ms. Early diastolic velocity (e') at the septal mitral annulus (**center left**) is reduced with age at 3.9 cm/s (average of 3 beats), and the septal E/e' ratio is 23. The inferior vena cava (IVC) diameter is 2.0 cm, with complete collapse with sniffing, consistent with normal mean right atrial pressure (RAP; **center right**). Peak tricuspid regurgitation (TR) velocity is increased to 3.7 m/s (**right**); therefore, the estimated pulmonary artery (PA) systolic pressure is 58 mm Hg. **B**, Step-by-step approach for applying 2025 American Society of Echocardiography guidelines. Based on the findings shown in Figure 6A, the mean left atrial pressure (LAP) is estimated to be elevated. Abnormal values are shown in red. BMI indicates body mass index.

It is good practice to check on the accuracy of the echocardiographic estimation of LAP against the invasive gold standard for patients with a PA catheter or those undergoing cardiac catheterization, and we often do that in our laboratory. Further, it is good practice to have a diastolic function assessment as part of the peer review of echocardiographic interpretation in QA sessions. This allows critique of the quality of the acquired data, correction of misunderstandings, and building consensus on how to perform the interpretation, such that the conclusion does not depend on who the interpreting physician is.

Diastolic Stress Testing

Diastolic exercise stress testing, usually with supine bike exercise, is performed in conjunction with transthoracic imaging. It is indicated in symptomatic patients where LAP at rest is normal or indeterminate. It can also be performed with treadmill exercise. During the supine bike, image acquisition is performed during exercise. During treadmill exercise, Doppler signals are obtained after termination of exercise and acquisition of 2-dimensional images. At baseline, mitral inflow at the level of the valve tips and mitral annulus tissue Doppler velocities, both by pulsed wave Doppler, are recorded. In addition, the peak velocity of TR by CW Doppler is obtained. Agitated saline should be administered as needed to obtain a complete TR envelope of the TR velocity. The same signals are acquired during exercise, and at 1 to 2 minutes after termination of exercise, since there can be partial or complete fusion of mitral inflow E and A velocities. The test is positive when peak average E/e' ratio ≥14 (or septal E/e' ratio ≥15 in case only septal velocities are recorded), and peak TR velocity ≥3.2 m/s.[26]

HFpEF Diagnosis

HFpEF diagnosis relies on the presence of symptoms and signs of heart failure, normal LV EF ≥50%, and LV EDV index that is ≤97 mL/m^2, in addition to findings of abnormal LV diastolic function as ascertained by invasive or noninvasive criteria.[30–32] There is a need for imaging data for this objective, given the criteria for HFpEF diagnosis noted above, and the limited accuracy of clinical findings, chest X-ray results, and natriuretic peptides levels when compared against the invasive gold standard.[33]

The 2025 ASE guidelines recommend a stepwise approach that aims at excluding diseases that mimic HFpEF as a first step, then determining LAP at rest based on the algorithm recommended above, and finally, in case of equivocal findings, diastolic stress testing by echocardiography or right heart catheterization. A recent study showed that the 2025 guidelines provide incremental value to the H2FPEF score in patients with intermediate probability and in patients with high probability for HFpEF, but not in patients with low probability.[34,35] The performance of comprehensive echocardiography in patients with intermediate probability is of particular interest as numerous patients under evaluation for HFpEF have an intermediate disease probability, which would trigger stress testing as the next step in their evaluation. Accordingly, a successful diagnosis of HFpEF based on echocardiography can decrease the number of patients referred for additional testing, be it with exercise echocardiography or right heart catheterization. LARS has a stronger relation with LV filling pressure in patients with depressed EF, but lower accuracy in patients with normal EF.[36,37] In comparison, recent studies have shown that noninvasive LA chamber stiffness, average E/e' ratio divided by LARS,[38–40] has incremental value and has the highest accuracy of any individual echocardiographic measurement.[35,41,42]

Assessment of LV Diastolic Function in Patients With Aortic Valve Disease

The usual approach in patients without aortic stenosis or aortic regurgitation is applied for assessment of LV diastolic function and estimation of LAP. This relies on mitral inflow velocities, e' velocities, E/e' ratio, peak TR velocity, and LA maximum volume index.[26,43] Importantly, echocardiographic stages of DD are associated not only with LV structural changes of interstitial and replacement fibrosis by CMR, but also outcome events of total mortality and heart failure hospitalizations in patients with at least moderate AR as assessed by CMR.[44] Interestingly, DD occurs in patients with significant AR irrespective of symptomatic status and is associated with outcome events in patients with New York Heart Association class I status. This exposes the need to evaluate the role of LV diastolic function in the timing of surgery for asymptomatic patients with significant AR.[45]

Assessment of LV Diastolic Function in Patients With MR

LV DD is present not only in patients with secondary MR, but also in patients with primary MR. Patients with longer LV time constant of LV relaxation, and higher LV chamber stiffness as determined by conductance catheters and pressure volume loops are left with higher LAPs and worse symptomatic status after successful mitral transcatheter edge to edge repair.[46]

The 2025 guidelines have the same criteria for diagnosing LV DD and estimating LAP in patients with MR as those in the 2016 guidelines.[26] For patients with ventricular functional MR, average E/e' ratio[47] and peak TR velocity are recommended. For patients with primary MR, pulmonary vein atrial velocity duration and the time difference between its duration and that of mitral A velocity at the level of the mitral annulus are recommended for drawing conclusions about LV EDP.[24,26,48] The time delay between the onset of mitral inflow and the onset of e' velocity relates to LV relaxation,[20,21,49] and when combined with IVRT can be used to estimate LAP.[49] A recent study showed that these indices of LV diastolic function are associated with LV structural changes of interstitial and replacement fibrosis by CMR, and outcome events of total mortality, heart failure hospitalizations, and surgery in patients with at least moderate primary MR as assessed by CMR.[22]

LAP can be estimated after mitral transcatheter edge to edge repair procedures (and in patients with patent foramen ovale) as well, since post mitral transcatheter edge to edge repair, there is iatrogenic interatrial communication with left-to-right shunt. It is possible to reach good alignment with this flow from the subcostal view. Using the modified Bernoulli expression, it is possible to estimate LA V pressure using the peak velocity at end systole and LA A pressure using the peak velocity at end diastole.[26] For either pressure, $LAP-RAP=4v^2$ (where versus is the peak velocity in m/s; Figure S2). Therefore, LAP can be given by $RAP+4v^2$.

Assessment of LV Diastolic Function in Patients With Mitral Annular Calcification

There is limited data on the accuracy of noninvasive estimation of LV filling pressures in patients with moderate or severe mitral annular calcification using invasive LV pressure as the gold standard.[50] For patients in sinus rhythm, the 2025 guidelines recommend starting with the E/A ratio. When the ratio is <0.8, LAP is usually normal, but LAP is elevated when the ratio is >1.8. When E/A is 0.8 to 1.8, IVRT should be measured. LAP is normal when IVRT is ≥80 ms, whereas LAP is elevated if IVRT <80 ms.

Assessment of LV Diastolic Function in Patients With Atrial Fibrillation

Based on the results of a multicenter study,[51] the 2025 ASE guidelines recommend a stepwise approach. In the first step, peak E velocity, DT, PASP or peak TR velocity when RAP cannot be estimated, and septal E/e' ratio are evaluated. When 3 or all 4 variables (E≥100 cm/s, DT ≤160 ms, septal E/e' >11, TR >2.8 or PASP >35 mm Hg) meet the cutoff values, then LAP is elevated (Figure 4). When no or only 1 abnormal value is present, then LAP is normal. In the presence of only 2 abnormal values, LARS, pulmonary vein systolic velocity to diastolic velocity ratio, and BMI are examined. When 2 of 3 or 3 of 3 cutoff values (LARS <18%, S/d ratio <1, BMI >30 kg/m^2) are present, LAP is elevated. Conversely, when all values fall short of the cutoff values, LAP is normal. When only 1 variable with suitable quality is available, then LAP is indeterminate. It should be noted that the yield of indeterminate LAP is low when all signals are acquired and are of good quality for analysis. As a semi-quantitative metric, variability of mitral inflow with varying cycle length can be examined, based on recording several cardiac cycles at a sweep speed of 50 mm/s. Patients with less beat-to-beat mitral inflow variability usually have elevated LV filling pressure.[52]

Estimation of LV Filling Pressure in Heart Transplant Recipients

Most heart transplant recipients show predominant early LV filling with normal LV diastolic function as they receive hearts from normal donors. Of note, mitral inflow by itself has limited accuracy in detecting the presence of LV DD in this patient population.[53] The recommended approach begins by obtaining the average E/e' ratio. An average E/e' ratio <7 supports the conclusion that LAP is normal, whereas a ratio >14 is seen with elevated LAP. When the E/e' ratio is 7 to 14, strain rate during the isovolumic relaxation period (SR_{IVR}), from all 3 apical views, is measured. A ratio of mitral E velocity to SR_{IVR} ≤200 cm is consistent with normal LAP, whereas a ratio >200 cm supports the conclusion of elevated LAP. Conversely, peak TR velocity is evaluated after obtaining the average E/e' ratio. A peak TR velocity ≤2.8 m/s is consistent with normal LAP, and a peak TR >2.8 m/s supports the conclusion of elevated LAP.[54]

Estimation of LAP in Patients With PH

Patients with noncardiac PH usually have normal or reduced LV and LA volumes and an impaired relaxation mitral inflow pattern. Patients with group II PH can have an enlarged LA and evidence of LV structural pathology as increased LV mass. Mitral inflow is often pseudonormal or restrictive.[55] The recommended algorithm in patients with PH begins with mitral E/A. A ratio ≤0.8 supports the diagnosis of noncardiac PH, and a ratio ≥2 is consistent with group II PH. When E/A >0.8 but <2, a lateral E/e' >13, or LARS <18% are consistent with elevated LAP.[56]

THE FUTURE

In the past few years, novel indices as well as new approaches relying on AI have been developed. The novel indices include comparing the timing of mitral valve opening with that of tricuspid valve (TV) opening, measuring LA conduit volume, measuring shear wave propagation velocity, and using microbubbles and subharmonic-aided pressure estimation.

Estimation of Intracardiac Pressure Using Subharmonic Aided Pressure Estimation

The ultrasound-enhancing agents, after intravenous injection, move throughout the cardiovascular system, including the cardiac chambers. When exposed to high insonation pressures, the microbubbles emit oscillations with frequencies ranging from subharmonic (frequencies below the fundamental frequency) to ultraharmonics. The amplitude of the subharmonic signals has a strong and inverse linear relationship with the surrounding pressure. Accordingly, it is possible to provide an estimate of pressure based on the signal amplitude for subharmonic frequencies. In one study, the relation between subharmonic-derived pressure and LV EDP by cardiac catheterization in 26 patients was excellent, with a mean difference between noninvasive and invasive LV EDP of 0.2 mm Hg and an SD of 2.2 mm Hg.[57] Although promising, this was a single-center study with a small number of patients. Careful attention to technique and analysis is critical to obtain high-quality data. Aside from the question of whether this technique can be successfully obtained in most clinical laboratories, large beat-to-beat variation of the amplitude of the subharmonic signals versus time was present, which raises the question of the impact of beat selection on the results.[58] Further, the authors used a large amount of ultrasound-enhancing agent, which comes with extra costs and the need for intravenous access. Thus, additional studies are needed to address these critical methodological concerns and show the feasibility of successful protocol execution in multicenter studies.

Shear Wave Propagation Velocity

Shear waves are produced by applying a short push of focused ultrasound, referred to as acoustic radiation force. The shear waves have a frequency in the kHz range. Thus, a very high frame rate is needed for image acquisition, in the thousands range. In most studies published on this technology, the velocity of shear waves was obtained using signals from the anterior septum in the parasternal views. In addition to shear waves induced by acoustic radiation force, naturally occurring shear waves result from atrial contraction, mitral valve closure (MVC), and aortic valve closure. It is possible to draw inferences about myocardial stiffness based on shear wave propagation velocity, since the velocity =√shear modulus/√density, and increased tissue stiffness leads to increased shear modulus.[59]

Earlier studies showed an increase in naturally produced shear wave propagation velocity with age and in patients with cardiac amyloidosis[60] (Figure 5). Further, a progressive increase was seen in the propagation velocity as DD grade increased from 1 to 3, along with a significant direct linear relationship between the shear wave propagation velocity and E/e' ratio.[60] In heart transplant recipients, intrinsic shear wave propagation velocity was significantly related to T1 relaxation time by CMR, and to mean wedge pressure obtained by right heart catheterization.[61] Most recently, LVEDP was measured in 84 patients undergoing left heart catheterization, and natural shear wave propagation velocity due to MVC was recorded with image acquisition obtained at an average frame rate of 1050±220 Hz.[62] Shear wave velocity was measured in the anteroseptal wall. The authors reported that the European Association of Cardiovascular Imaging algorithm[32] for the estimation of LV pre-A pressure had an AUC of 0.78 for detecting elevated LVEDP. The shear wave propagation velocity at MVC had a significant relation with LVEDP (r=0.71; P<0.001) and a similar AUC for detecting elevated LV pre-A pressure as that of the European Association of Cardiovascular Imaging guidelines (AUC: 0.79; sensitivity: 87%; specificity: 65%). In 16 patients with decompensated heart failure, a significant decrease in shear wave velocity was observed after diuretic therapy (7.1±1.7 m/s versus 5.3±1.3 m/s; P<0.002).

Although these results are promising, it is important to note that propagation velocity depends not only on the tissue properties but also on LV wall thickness and loading conditions. Another limitation is obtaining the measurement from 1 region, which may not be reflective of other segments in the presence of segmental

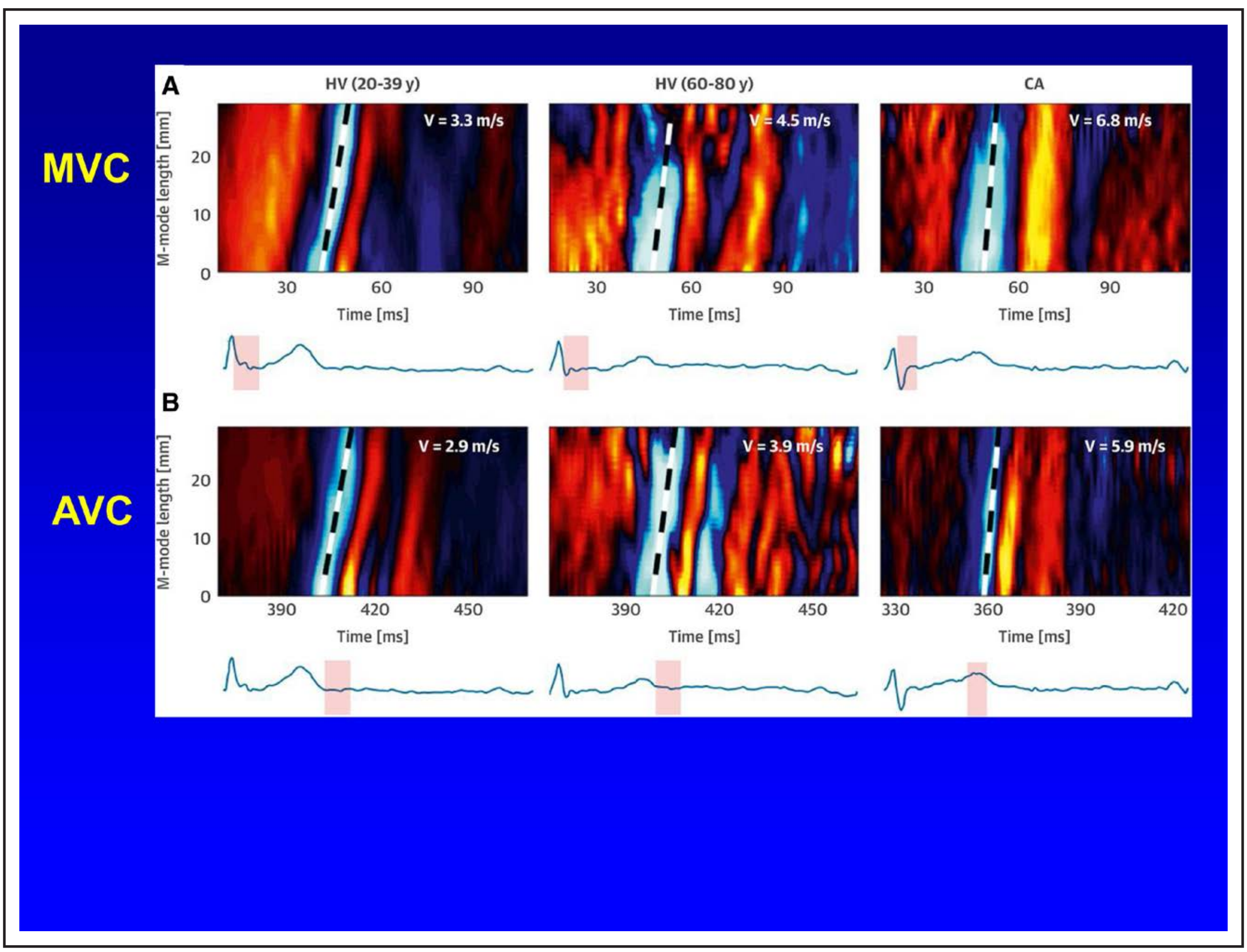

Figure 5. Shear wave velocities in normal subjects and patients with cardiac amyloidosis.
Shear waves after mitral valve closure (MVC; **A**), and aortic valve closure (AVC; **B**) from a normal subject (healthy volunteer [HV]) with age between 20 and 39 years (**left panel**), another normal subject with age between 60 and 80 years (**middle panel**), and a patient with cardiac amyloidosis or CA (**right panel**). The highlighted region of the EKG indicates the time interval covered by the M-mode map in the corresponding part of the figure stands for the shear wave propagation velocity. Notice the higher velocity in the older subject and the highest velocity seen with cardiac amyloidosis. Reproduced from Petrescu et al[60] with permission. Copyright ©2019, Elsevier.

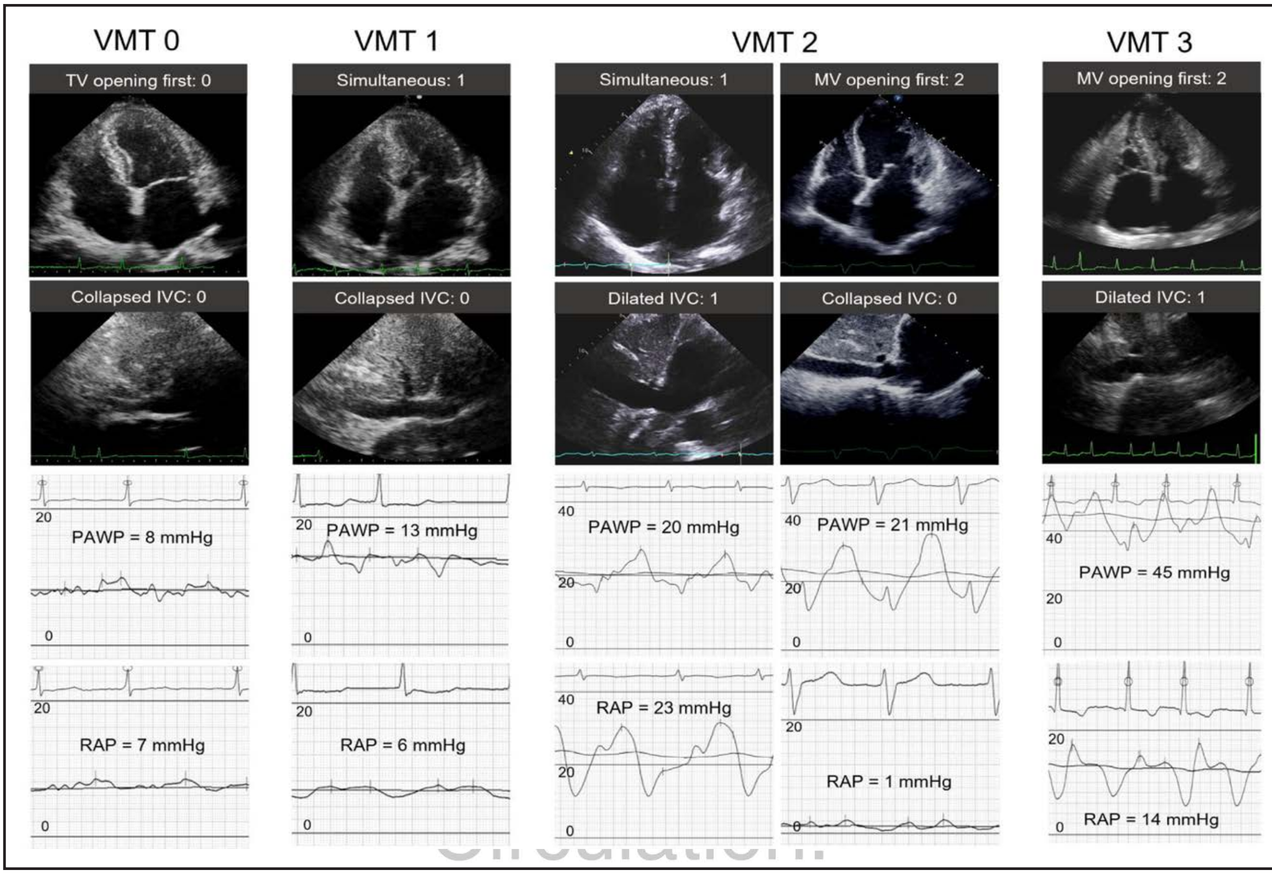

Figure 6. Order of opening of the tricuspid valve (TV) and mitral valve (MV) and estimation of left atrial pressure.
Apical 4-chamber views in early diastole, subcostal views, and corresponding pulmonary artery wedge pressure (PAWP) and right atrial pressure (RAP) recordings are presented. The order of MV and TV opening was visually assessed in the apical 4-chamber view and scored as follows: 0=TV opening first, 1=simultaneous opening, and 2=MV opening first. When the inferior vena cava (IVC) diameter was >2.1 cm and collapsed by <20% during normal respiration, 1 point was added. The VMT (time difference between mitral valve and tricuspid valve opening) score is calculated in 4 grades from 0 to 3. Reproduced from Nishino et al[67] with permission. Copyright ©2024, Wolters Kluwer Health, Inc.

differences in structure and function. Further, shear waves produced by MVC propagate during systole and thus are not necessarily reflective of only LV diastolic properties.

For the technique to have a wide clinical application, it should have high feasibility and a narrow range for normal values. These 2 aspects were examined in 239 asymptomatic subjects from a community-based heart failure surveillance program.[63] The shear waves produced by atrial contraction, MVC, and aortic valve closure were recorded from the parasternal and apical views. Subjects were classified into normal and abnormal groups based on clinical and echocardiographic findings. Manual measurements were more frequently obtained in comparison with automated measurements. Feasibility ranged from 62% to 91% based on the origin of the shear waves and the view from which they were recorded. Surprisingly, clinical and echocardiographic indicators of hemodynamics showed weak or no significant correlation with the shear wave propagation velocity. In addition, the agreement was not good for the shear wave velocity measurement emanating from MVC when the signals acquired from the parasternal and from the apical views were compared. Shear wave propagation velocity for shear waves produced by MVC and by aortic valve closure was not different between subjects with and subjects without cardiovascular disease, but the velocity of shear waves generated by atrial contraction was significantly different. While there was little bias, the limits of agreement were wide for all measurements. The wide range of normal values and the normal values seen in patients with subclinical disease raise concerns about the ultimate clinical application of using these signals for early diagnosis of cardiac pathology. While the existing studies address the value of this technology by itself, shear wave propagation velocity may also be combined with other echocardiographic indices for a more refined assessment of LV diastolic function.

LA Conduit Volume

LV filling depends on the inflow of blood from the LA in early diastole (passive filling) due to LV relaxation that causes low LV early diastolic pressure with a positive LA-LV pressure gradient for blood flow into the ventricle. Another phase of LV filling occurs in late diastole, related to LA contraction, increasing LAP, and, in turn, the LA-LV pressure gradient. For adequate LV filling, the LA reservoir and contractile functions have to be preserved. In the early phases of DD with impaired LV relaxation, LV filling in early diastole is reduced, and filling becomes dependent on LA contraction. With the progression of DD and increased early and late LV diastolic pressures, LA contraction is reduced because of increased LA afterload. At this stage, LV filling becomes most dependent on the LA conduit volume, which is the volume of blood that passes from the pulmonary veins through the LA and into the LV. It can be measured as the difference between LV stroke volume and LA emptying volume. In turn, the LA emptying volume is obtained as the difference between the LA maximum and minimum volumes.

One study noticed an increase in LA conduit volume with progressive stages of DD assessed by echocardiography. In that study, the conduit volume was measured by multiphasic cardiac CT scanning.[64] However, the conduit volume can also be measured by echocardiography and by CMR.[65] While promising, additional studies are needed to validate the association of LA conduit volume with invasive LV filling pressures as well as with heart failure hospitalizations. In addition, research studies should aim to evaluate their incremental value over stand-alone conventional measurements and over existing algorithms for LAP estimation.

Timing of MV and TV Opening

In normal hearts, TV opening precedes MV opening. Therefore, by comparing the timing of opening of the 2 valves, LAP can be estimated, albeit as a binary outcome of normal versus elevated LAP. Expectedly, patients with earlier opening of the MV before TV opening, despite an elevated RAP, have higher LAP than those where earlier MV opening occurs in the setting of a normal RAP. A scoring system can be applied, taking into account RAP and the timing of MV opening relative to TV opening. This concept was validated in patients in sinus rhythm,[66] in patients with atrial fibrillation[67] and was associated with heart failure hospitalizations in patients with HFpEF.[68] Figure 6 shows the scoring system application in patients with atrial fibrillation. Although there is a solid conceptual basis for this approach, its accuracy remains to be examined in a multicenter study, as well as its incremental value over the existing methodology. Further, it can be challenging to apply to patients with atrial fibrillation and a large variation in RR interval.

Generation of LV Diastolic Pressure Curve Using Echocardiographic Imaging

A recent study reported on estimating LV minimum pressure, pre-A pressure, and LVEDP, along with reconstruction of the pressure curve in patients (indication for catheterization was evaluation for CAD) undergoing cardiac catheterization, where LV pressures were obtained using high-fidelity pressure catheters.[69] LV minimum pressure was estimated based on a regression equation that included LV volumes and strain, LARS, mitral inflow velocities, systolic blood pressure, and body mass index. Pre-A pressure was obtained as the sum of the minimum pressure and maximal transmitral pressure gradient. Transmitral pressure gradient was derived by the simplified Navier-Stokes equation. The regression equation was obtained based on data from 81 patients, and the method was validated in a separate group of 19 patients. LV diastolic pressure curves were derived by adjusting a reference curve based on pressure estimates at the above specific points during diastole. The authors reported good agreement between measured and estimated LV diastolic pressure curves. Although promising, additional studies are needed to evaluate the accuracy of the method in a multicenter setting and in patients with a wide spectrum of cardiovascular diseases.

Application of AI to Diagnose DD and to Estimate LA Pressure

There are several approaches that have been evaluated for AI assessment of LV diastolic function.[26] Machine learning methods have been used for view classification, segmentation, and obtaining 2-dimensional and Doppler measurements. Further, there are approaches based on drawing conclusions from multiple views or a single apical 4-chamber view.[70–74] The measurements relied on include LV and LA volumes, mitral inflow, tissue Doppler velocities, and LA and LV longitudinal strain. The data, once obtained, can be handled as a rule-based decision tree algorithm. Alternatively, deep-learning neural networks to train models for diagnosing and grading DD, and for diagnosing HFpEF have been developed. Some models used diastolic function grading based on the 2016 ASE/European Association of Cardiovascular Imaging guidelines, and thus an indeterminate grade of diastolic function is one of the possible outputs from these models. Others relied on classifying diastolic function based on outcomes, which led to a reduction of indeterminate cases.[75,76]

While the DL approach is an important new development, there are challenges to its successful application in clinical practice. A recent study[77] reported significant variation in AUC for LV EF measurement with implications for heart failure management based on the results of DL models for EF derivation. AUC ranged between 0.71 and 0.98 based only on the characteristics of the training sample. Furthermore, model performance was worse when applied to an unrelated external data set. Validation of AI models against invasive pressure measurements and clinical events is needed to establish the robustness of AI models and enhance their adoption in day-to-day clinical applications.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Figures S1–S2

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